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Integrated approach to Wire Arc Additive Manufacturing (WAAM) optimization: Harnessing the synergy of process parameters and deposition strategies

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ABSTRACT

The flexibility of Additive Manufacturing (AM) technologies in the metal 3D printing process has gained significant attention in research and industry, which allows for fabricating complicated and intricate Near-Net-Shape (NNS) geometry designs. The achievement of desired characteristics in Wire-Arc Additive Manufactured (WAAM) components is primarily contingent upon the careful selection and precise control of significant processing variables, including bead deposition strategy, wire materials, type of heat source, wire feed speed, and the application of shielding gas. As a result, optimizing these most significant process parameters has improved, producing higher-quality WAAM-manufactured components. Consequently, this has contributed to the overall rise in the method's popularity and many applications. This article aims to provide an overview of the wire deposition strategy and the optimization of process parameters in WAAM. The optimization of numerous wire deposition techniques and process parameters in the WAAM method, which is required to manufacture high-quality additively manufactured metal parts, is summarised. The WAAM optimization algorithm, in addition to anticipate technological developments, has been proposed. Subsequently, a discussion ensues regarding the potential for WAAM optimization within the swiftly growing domain of WAAM. In the end, conclusions have been derived from the reviewed research work.

1. Introduction

Additive manufacturing (AM) has been widely discussed by engineers and researchers as a way to lead the manufacturing industry in producing medium-to-large scale Near-Net-Shape (NNS) components more quickly, economically, and with less waste [1]. Due to the multi-purpose capabilities within AM, many manufacturing industries have shifted towards this technology instead of subtractive manufacturing (SM) [1,2]. Furthermore, AM can manufacture relatively complex geometries better than SM due to how it is revolutionized for various types of materials [3]. AM of NNS metal components comprises of powder-based and wire-based technology [1]. Fig. 1 show types of AM process based on the raw materials used and their respective technologies found throughout recent research developments.

AM techniques can be classified into powder-based and wire-based processes based on the type of energy source utilised for feedstock distributions [4–7]. Powder-based additive manufacturing offers benefits such as a relatively small pool, a concentrated energy source, and precise depositing. Heterogeneous metallic components with complicated geometries can be manufactured by adjusting the powder composition proportions of the feedstock materials in an immediate fashion. This ability demonstrates the wide-ranging practical utility of powder-based

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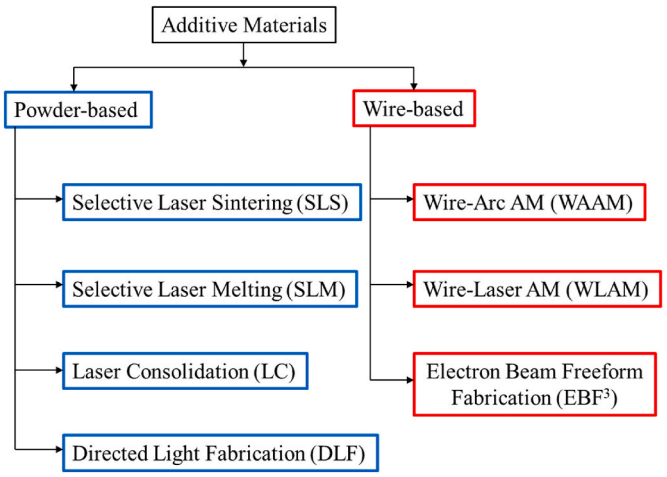


Fig. 1. Additive materials classification based on type of raw materials and respective technologies.

additive manufacturing technology, encompassing applicability in the biomedical, oil and gas fields. Challenges in achieving dense components include the substantial expense of powder materials, limited depositing efficiency, low utilization rates, porosity and lack of fusion [8–11].

The wire extracted from the feedstock is introduced promptly into the molten pool produced by the heat source, enabling the wire material to melt either in a shielding gas environment or in a vacuum state in a layer-by-layer fashion in wire-based additive manufacturing [12]. Wire-based additive manufacturing methods can produce wholeheartedly dense parts or components with precise control over microstructural characteristics [5,13,14]. Wire-based additive manufacturing methods have enabled the cost-effective production of profound metallic parts with moderate complexity [15]. The primary limitations of the Wire-based Additive Manufacturing technology are low precision resolution and an average surface finish. Components manufactured through this process require finishing on the surface using CNC machines [16,17].

WAAM is one of the most prevalent AM processes for 3D metal printing and offers several advantages. WAAM is more effective at making large-scale components than other AM technologies because it uses low-cost materials and is efficient and productive [18]. Additionally, metal-wire has many advantages over powder-feed technology. The cost of metal wire is around 10% of metal powder. Thus, the WAAM method performs well with high deposition rate at a cheap cost, making it ideal for fabricating large components with an acceptable buy-to-fly ratio from costly metals [19,20]. Due to powder-feed technology's poor deposition rate of 10 g/min [21], wire-feed is favoured and widely employed in the industry. Table 1 summarizes some of the WAAM process applications concerning the welding heat source employed by researchers.

There are four primary heat sources commonly used for the WAAM process to melt the wire metals: gas metal arc welding (GMAW), cold metal transfer (CMT), gas tungsten arc welding (GTAW), and plasma arc welding (PAW). GMAW-based WAAM is known for its simple tool path-planning algorithm, user-friendliness, cheaper capital cost, and greater convenience to use than other heat sources [28]. The CMT-based WAAM technique is frequently utilised compared to other WAAM processes due to its high deposition rate, low heat input, little dilution of base metal alloys, controllable structural deformation, and low residual stress accumulation [60–62]. GTAW and PAW-based WAAM use the same method of producing molten pools using non-consumable tungsten electrodes. Unlike GMAW, GTAW and PAW require an external wire feed machine or separate wire feeding system to supply the feed materials [63].

Table 1
Applications of WAAM technology.

Application	Welding Heat Source			
	GMAW	CMT	GTAW	PAW
Aerospace	[22–26]	[27–30]	[31–35]	[23,36,37]
Automotive	[38,39]	[40–42]	[43,44]	–
Marine	[45,46]	[47,48]	–	–
High Temperature	[49,50]	–	[51]	[52]
Corrosion Resistivity	[53–56]	[57]	[58,59]	–

In WAAM process, Wire Feed Speed has a very strong correlation with the speed of deposition rate, which significantly affects the overall quality of the manufactured parts [64]. In addition, according to the research conducted by B. Wu et al. [64], it is possible to achieve consistent and stable metal transfer behaviour in the WAAM process by meticulously controlling the inter-pass temperature and using appropriate localised gas shielding. Meanwhile, Dinovitzer et al. [65] state that the main cause of wire-feeding AM process problems significantly originated from residual stress and distortion formation. It revealed that the WAAM process is a viable wire-feed technology as long as optimal values were selected for the travel speed, wire feed rate, current, and gas flow rate. Controlling process parameters in WAAM has proven difficult to optimize and sustain efficiency [66]. This requires the development of an approach to optimize the manufacture of big near-net shape (NNS) parts by removing or minimising errors in the final product.

Fig. 2 presents the research challenges and future directions for the WAAM process. The deposition strategy and process parameters are crucial in the widespread adoption of WAAM as a high deposition rate and have become the main focus of this review paper. Currently, many methods are employed for optimization, particularly for WAAM process. Due to the very intricate nature of WAAM, it is necessary to delve into diverse topics, including process optimization, deposition strategy, and adaptability of statistical tools.

Several scholarly review articles on the WAAM have been written by researchers, focusing on advanced systems, design, utilization, real-time monitoring, sensing, and process control [16,28,67–69]. Research progress has been reported in the field of WAAM process parameters optimization [57,60,65,70] and deposition strategy optimization [56,57,66,71–75]. Researchers have not extensively covered the review domain of optimizing WAAM processing parameters and deposition strategy in their review articles [69,76–80]. Hence, collecting and reviewing thorough data from published articles in the optimization technique and deposition strategy fields is imperative, as depicted in Fig. 3.

This article aims to review the status of recent research and development in the field of optimization techniques for the WAAM process. A comprehensive discussion is conducted on the process parameters for optimizing the WAAM process. A comprehensive analysis of several slicing and path planning strategies used in WAAM weld bead deposition is discussed. Furthermore, the optimization methods researchers employ to optimize their findings, emphasizing statistical analytic techniques, are presented. A proposed method for optimizing WAAM processes and future perspectives in process optimization are presented, followed by concluding remarks.

2. Process parameters optimization

To fully optimize the utilization of a manufacturing process, an individual must be able to identify all the process parameters associated with that specific process. In addition, by prioritising optimization, it becomes possible to achieve both simplicity of process implementation and enhanced user experience. Hence, in order to obtain NNS components of superior quality and without any defects in the Wire Arc Additive Manufacturing (WAAM) process, a meticulous selection and implementation of a combination of process parameters such as wire

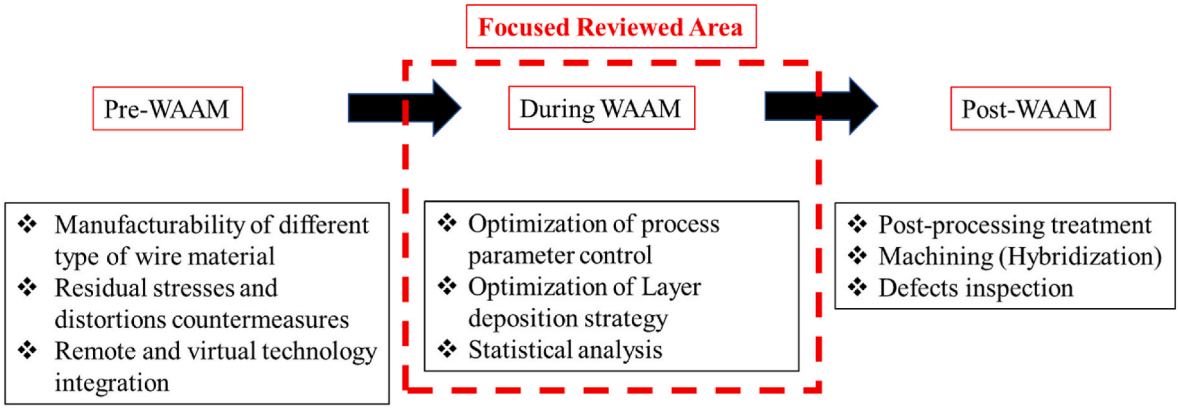


Fig. 2. WAAM difficult research hotspot.

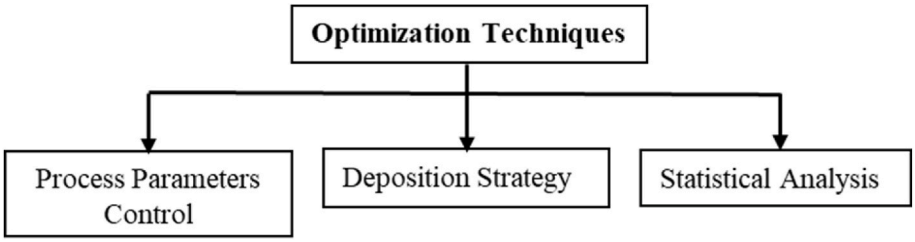


Fig. 3. Optimization technique focused research area.

diameter, wire feed rate (WFR), travel speed, welding voltage, welding current, nozzle-to-work distance (NTWD), shielding gas torch angle, and stepover distance are necessary. This section explicitly addresses the critical process parameters that significantly impact the properties of NNS components manufactured by WAAM.

2.1. Wire feed rate

The primary issue that WAAM has encountered is deciding whether to enhance the weld bead deposition rate or decrease the heat input distribution. This is because Wire Feed Rate (WFR), also referred to as wire feed speed, is closely correlate with the necessary heat input [81–84]. Hence, choosing the optimum WFR is essential to ensure that excessive slowness does not compromise the residual stress accumulated in the deposited metals. Conversely, employing rapidity might result in deformation during the formation of each layer of beads. The deposition process will generate excessive heat, leading to the remelting of previously created layers and ultimately causing microstructure and bead shape degradation. Choosing an appropriate WFR can enhance the quality of surface evenness when a low deposition rate is required. Furthermore, studies by Baffa et al. [84] have demonstrated that WFR significantly affects the appearance of ER70S-6 steel weld bead using GMAW, as shown in Fig. 4. Various WFR variations result in high-quality bead dimensions.

The manipulation and control of WFR were also affected by the heat source selected. The WAAM method, which utilises the CMT technique, employs a controlled low-heat input and low WFR to provide a consistent deposition surface structure. This effectively prevents the weld pool

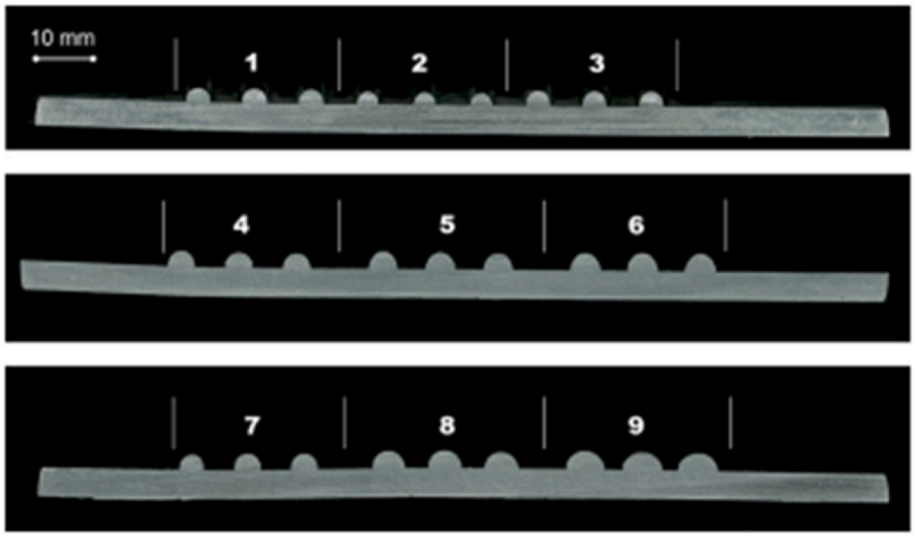


Fig. 4. Cross-sectional observation of bead dimensions using nine different WFR [60].

from overflowing or collapsing [85]. In order to get the optimal deposit quality and deposition rate, WFR is also essential [86]. In addition, increasing the WFR and energy input by giving a more significant current has been proven to be an effective method for enhancing the deposition rate [87]. The viability of the deposition technique for optimizing weld bead morphology has been established by considering WFR, welding current, travel speed (TS), substrate tilt angle, and welding torch angle [88].

Xiong et al. [89] discovered that the inclination angle of H08Mn2Si steel components is inversely proportional to the WFR values used, demonstrating the importance of WFR manipulation in optimizing the heat input supplied through GMAW. In addition, the extent of design geometry complexity for overhang features, circular patterns, and angular structures of UTP 759 Ni–Cr–Mo alloy was optimally controlled by a combination of factors, including type of heat source, current, voltage, and WFR, as shown in Fig. 5 [90]. Lower WFR values were used for non-vertical material deposition to ensure appropriate heat input, promoting adequate adhesion of the material on each succeeding layer while accounting for the effects of gravity. This demonstrated the significance of the WFR variation in influencing the ideal geometric characteristics of the component manufactured by WAAM.

Furthermore, WFR also impacts the development of welding bead characteristics, including bead height (BH), bead width (BW), and contact angle (θ). The influence of wire feed speed on bead width is challenging to control, as illustrated in Fig. 6(a). Initially, the bead width deposited reaches its maximum value but subsequently begins to diminish as the wire feed value increases. Conversely, the contact angle continues to increase with the wire feed speed, as illustrated in Fig. 6(b), suggesting that inadequate wetting has a detrimental effect on the workpiece [91]. Martina et al. [92] and C. Wang et al. [87] achieved a similar conclusion in their WAAM deposition technique for Ti–6Al–4V components using PAW heat source.

2.2. Travel speed

Along with heat input and WFR, travel speed is an important process parameter that should be precisely regulated since it substantially impacts the deposit progress, deposits geometry and thermal cycles of an adequate bead shape of H08Mn2Si steel using GMAW [93]. Travel speed is critical in achieving consistent layer-by-layer deposition in the WAAM process to manufacture NNS components with enhanced mechanical characteristics and intricate geometric designs. Higher deposition rates are possible at lower travel torch speeds, leading the welding arc to impact more on the molten weld pool (bead) than on the substrate material or the subsequently produced bead. Nevertheless, as the torch travel speed increases, the thermal energy per unit length of bead deposited will initially increase but then remain constant over time. Furthermore, an excessively high torch travel speed will result in inadequate metal supply to the deposited bead, as it diminishes the heat input distributed throughout the WAAM process, thereby influencing the melt-through depth [94]. As a result, selecting the optimum torch travel speed is essential to achieving components with improved integrity throughout the WAAM process.

According to Warsi et al. [82], arc power and travel speed are the most important criteria influencing the development of ER70S-6 steel fusion zone (FZ) geometry on a A36 steel substrate using a Computer Numerical Controlled Gas Metal Arc Welding (CNC GMAW). As travel speed rises gradually, less heat is emitted per unit length, causing the FZ's cross-sectional area to decrease. Furthermore, the geometrical characteristics of continuous single-layer aluminum deposits using CMT can be optimized by employing the optimum travel speed values with varying electric power, as denoted by P5, P6 and P7 in Fig. 7, having values of 1039 W, 1470 W, and 1666 W, respectively [95].

Travel speed was varied to identify the optimal structural integrity of components manufactured by WAAM. The findings indicate that when the electric arc travel speed increased, there was a decrease in the pores' sizes and volume fractions of each specimen, as seen in Fig. 8. This is highly beneficial for researchers aiming to reduce defects especially humping phenomenon in steel and aluminum components produced using the GMAW, GTAW, and CMT heat source [96–98].

Z. Wang et al. [99] proposed an improvement technique that combines varying travel speeds and an additional return path for each layer deposition in order to maximise high strength steel component structural integrity using CMT technique, as shown in Fig. 9. When maintaining a constant travel speed value from start to finish of a WAAM process, it is possible to deposit a bead; nevertheless, a compensation approach is required to deal with the bead's geometric irregularities challenges precisely.

Travel speed variations were frequently employed to improve the mechanical characteristics of a WAAM-manufactured component. According to research conducted by Yangfan et al. [100], the ultimate tensile strength (UTS) and yield strength (YS) of Inconel 625 increased as the travel speed increased, as shown in Fig. 10. This can be attributed to the precipitate phase that formed inside Inconel 625 when using CMT, causing a modification in grain structure and a reduction in the solid solution strengthening element.

2.3. Heat input

The arc current and voltage are other essential elements in the optimization process WAAM. They are the primary process control parameters that regulate the amount of heat input provided and dispersed, hence influencing the deposition of the weld bead. It is essential to prevent irregularly deposited beads and poor surface roughness. Additionally, an excessive influx of heat can result in the re-melting of previously deposited layers, adversely affecting the microstructure,

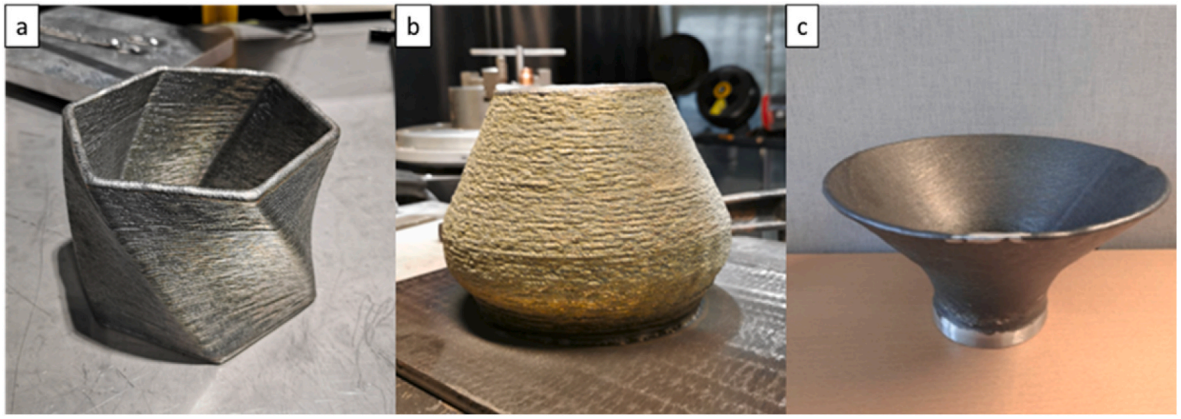


Fig. 5. The overhang WAAM structure: (a) twisting hexagon, (b) vase, and (c) bowl [90].

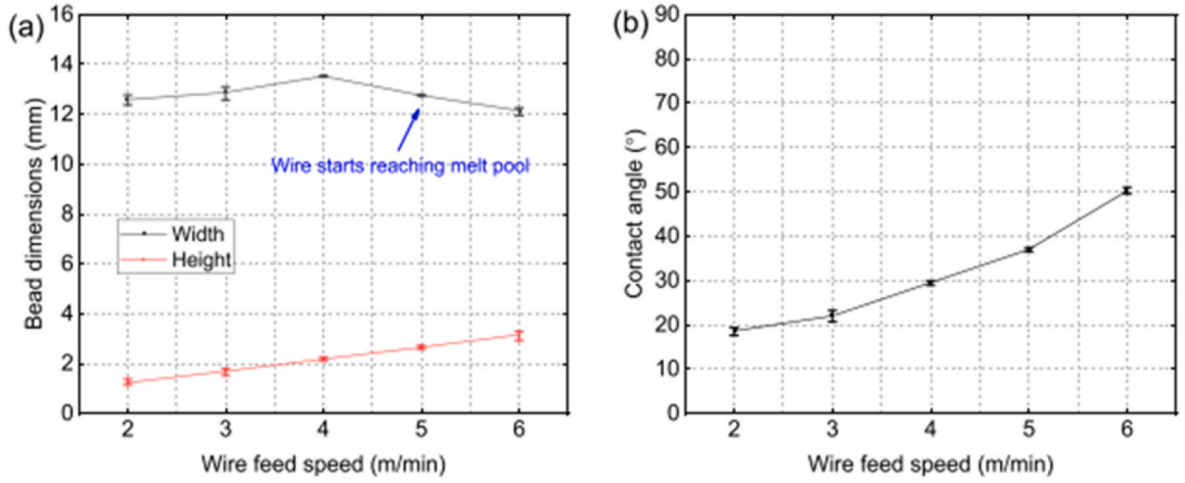


Fig. 6. The effect of wire feed speed on (a) bead dimensions, and (b) contact angle [91].

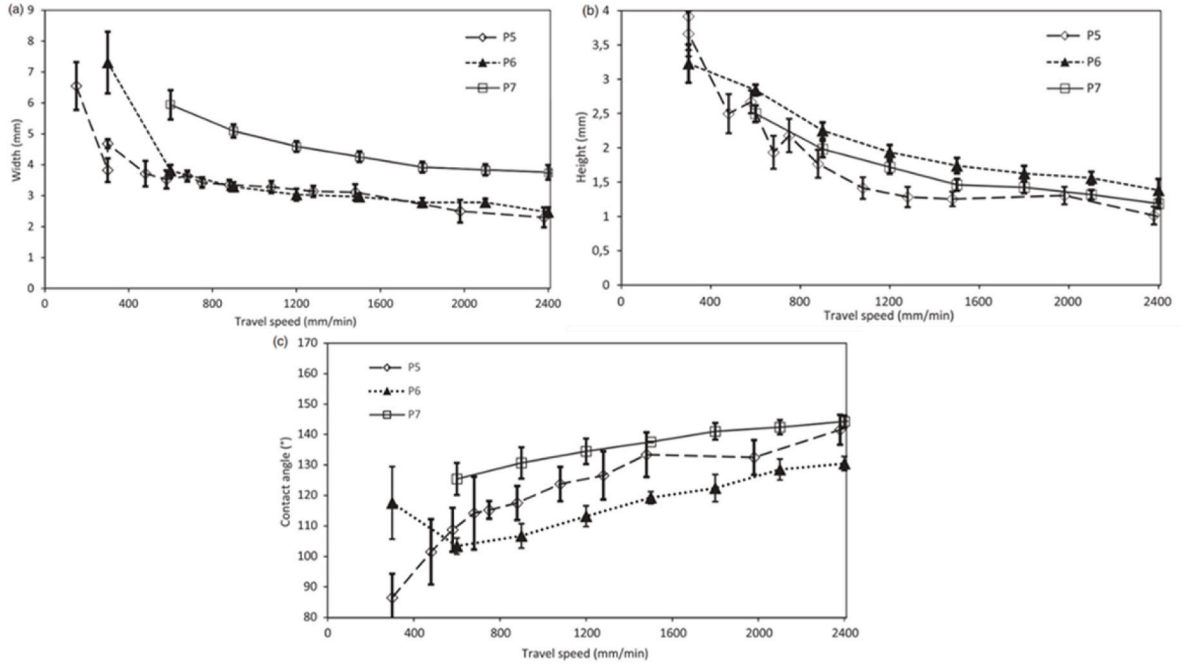


Fig. 7. Deposited bead dimensions (at varying electric power, denoted by P5, P6, P7) (a) width, (b) height, and (c) contact angle against the travel speed (mm/min) [95].

dimensions of the bead, and its mechanical characteristics. It is essential to optimize the current and voltage to enhance the overall efficiency of the process. Proper and steady regulation of heat input is critical to achieving the desired melting rate for metal deposition in WAAM [101]. The arc current, voltage, and travelling speed are the main factors that determine the appropriate thermal profile for WAAM. The heat input in the WAAM process, supplied by the heat source (GMAW, CMT, GTAW, PAW), refers to the amount of heat required to enable the deposition of metal on a substrate surface. This heat distribution is necessary to establish the required thermal gradient and ensure a proper heat cycle from melting to cooling to solidifying the deposition. The heat input is typically measured in joules per mm and derived from the subsequent equation (1) [102]:

$$Q = \frac{V\,(U) \times I\,(A)}{TS\,\left(\frac{mm}{min}\right)} \tag{1}$$

Where Q represents the heat input, V is voltage, I represents current, and TS is the travel speed of the torch.

$$Q = \eta\,\frac{V\,(U) \times I\,(A)}{TS\,\left(\frac{mm}{min}\right)} \tag{2}$$

The heat input calculation in the WAAM process has been modified by including the thermal efficiency coefficient (η) in equation (1). The value of η is influenced by the arrangement of the heat source in the WAAM process, the design of the deposition geometry, and the type of material used, as depicted in equation (2) [103–105]. This allows for the optimal amount of heat required to melt down the metals in the WAAM process. Thereby minimising defetcs such as undulated surface morphologies, inconsistent layer deposits, and time required for post-processing treatment.

Controlling interpass temperature and adhering to heat input standards are among effective methods for maintaining thermal boundary

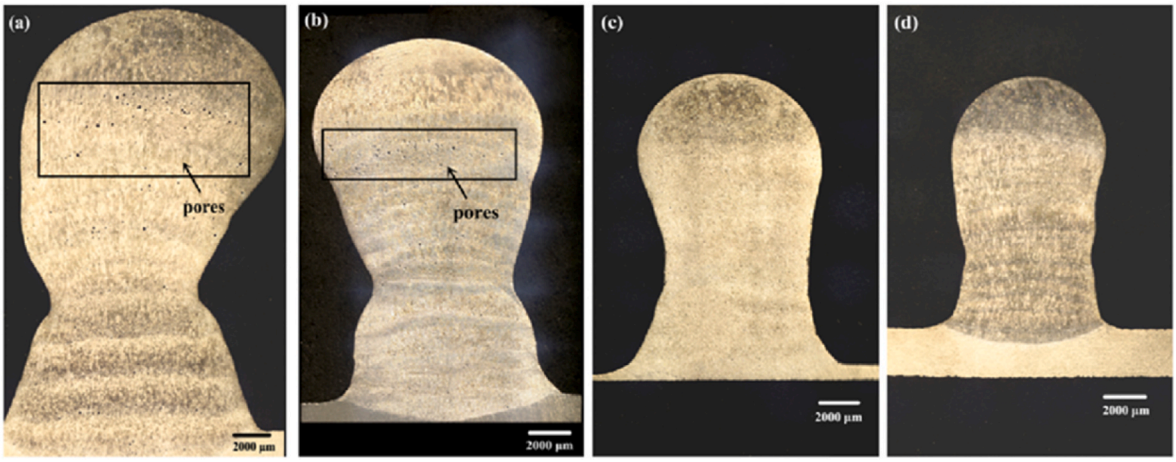


Fig. 8. Deposited metal's cross-sectional observation at various travel speed values: (a) 150 mm/min, (b) 250 mm/min, (c) 350 mm/min, and (d) 450 mm/min [96].

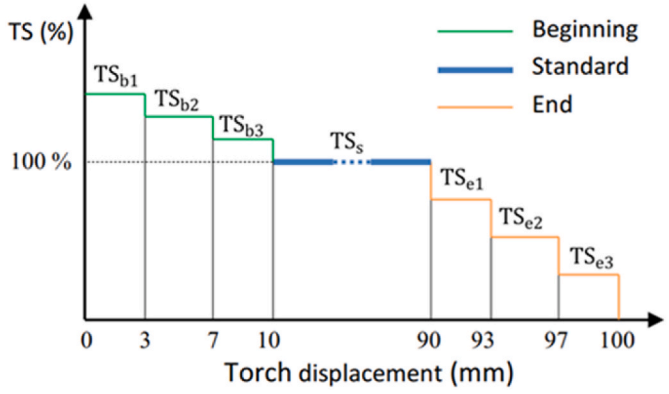


Fig. 9. Illustration of different travel speed configurations [99].

stability in WAAM process. Hence, interpass optimization seeks to minimise the temperature gradient at the fusion line between two layers. Meanwhile, it is essential to control heat input to eliminate the adverse effects of thermal disturbance resulting from heat accumulation [106]. There are numerous significant factors that have influenced the optimization of heat input in the WAAM process. Table 2 displays the correlation matrix among the factors that influence the optimization of heat input with regard to the performance of the manufactured WAAM product. The factor's influence on the performance of the WAAM product is designated as STRONG (S) if it is highly significant. At the same time, it is classified as WEAK (W) if it still has an impact but is not as significant. Based on the matrix table, the heat input, controlled by voltage and current, has the most significant impact on the overall performance of a WAAM product. Meanwhile, the structural integrity, which is determined by the Effective Wall Thickness (EWT) and Surface Waviness (SW), is greatly affected by each component that influences the optimum heat supplied.

According to Lervåg et al. [150], various material characteristics can be linked to the accomplishment of an optimized WAAM process through variation of heat input. These characteristics include tensile data, hardness value (HV), toughness, fracture morphology, microstructure characterisation, and the formation of intermetallic compounds. In addition, variations in heat input will substantially impact the distribution of optimum pore content and the sensitivity of the deposition process to hydrogen absorption [104]. Increased heat input and the use of various heat source modalities considerably contribute to the high abundance of pore numbers [151].

Furthermore, by utilising varying current (A) values, the %EP time cycle for each aluminum layer deposited can be optimized, leading to an optimal outcome since the cooling mechanisms operate linearly with GTAW heat input [152]. Controlling the quantity of heat input supplied is an additional method for optimizing the WAAM process, given that the phases of microstructure obtained are substantially correlated with the thermal cycles of each layer [153]. Additionally, weld bead geometry can be optimized by modulating the heat source of the WAAM process in order to improve control over process parameters. In their study, Greebmalai et al. [154] found that using double pulse current (DP-GMAW) for optimal heat input instead of single pulsed current

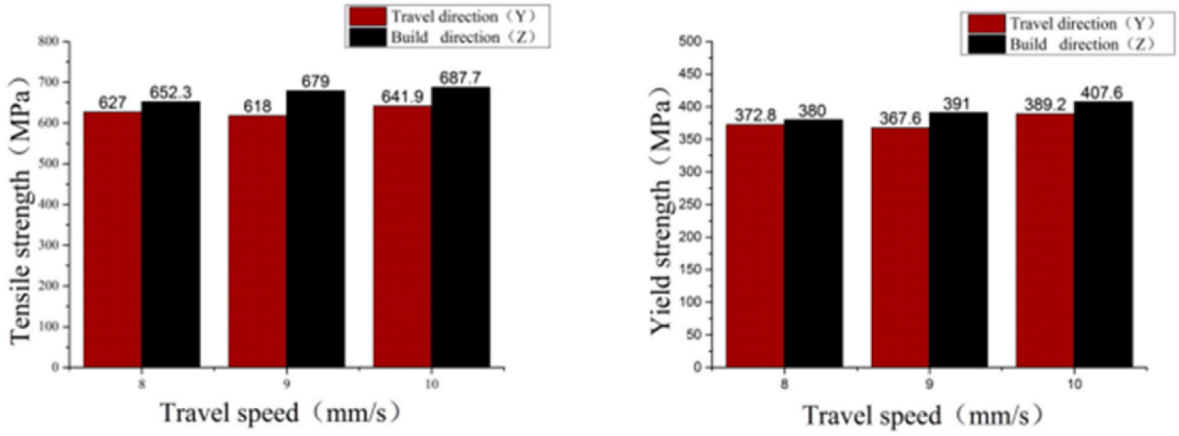


Fig. 10. (a) Ultimate tensile strength and (b) yield strength of the travel and build direction at various travel speeds [100].

Table 2
Relationship matrix between factors affecting the optimization of heat input with the finished product performance.[22,55–57,60,63,65,66,70,96–98,107–149]

(S = STRONG, W = WEAK, N = NO EFFECT)

Factors \ Performance	Structural Integrity	Surface Finish	Substrate warping	Residual Stress	Porosity	Cracks	Hump	Tensile Strength	Hardness	Micro-structure
Type of heat source	**S [107-110]**	**W [108]**	**N**	**N**	**S [66, 111-113]**	**N**	**S [97, 98]**	**N**	**W [114]**	**W [107]**
Type of wire material	**S [115]**	**S [116]**	**S [117]**	**W [118, 119]**	**S [66, 113, 120, 121]**	**S [66, 113, 122]**	**N**	**S [55, 123-129]**	**S [127, 130, 131]**	**S [56, 124, 127, 129, 132, 133]**
Wire Feed Speed	**S [57, 134]**	**S [57, 135]**	**N**	**S [65]**	**S [136]**	**S [57]**	**W [137]**	**S [60, 65]**	**S [60, 65]**	**S [57, 138]**
Travel Speed	**S [57, 137]**	**S [70, 135, 139]**	**N**	**N**	**W [140, 141]**	**S [70]**	**S [137, 142]**	**S [60, 65, 96, 139]**	**S [60, 65, 96, 139]**	**S [57, 96, 138, 139]**
Voltage (V)	**S [70, 143]**	**S [70]**	**S [144]**	**S [145, 146]**	**S [147]**	**S [70]**	**N**	**S [70]**	**S [70]**	**S [65, 147]**
Current (A)	**S [70, 105, 143, 148]**	**S [70, 135]**	**S [144]**	**S [145, 146]**	**S [147]**	**S [70]**	**S [22, 149]**	**S [70]**	**S [70]**	**S [65, 147]**

resulted in notable benefits in terms of the appearance of a single bead when using ER5356 aluminium alloy wire. Heat input can be optimized by using a by-pass current to improve deposition efficiency in an additive manufacturing (AM) system [155].

2.4. Shielding gas employed

Another crucial factor that needs careful consideration for optimization of WAAM process is the selection of shielding gas. This is due to the challenges associated with shielding gas employed and its influence on performance of the finished WAAM product. The welding torch setup provides the shielding gas in the majority of circumstances. Table 3 displays the correlation matrix between the factors that influence the optimization of shielding gas and the ultimate performance of the WAAM product, as reported in various studies. Based on Matrix Tables 3 and it is evident that fewer variables influence the end product's performance compared to Matrix Table 2.

However, the use of shield gas still significantly influences some characteristics, such as the structural integrity and mechanical behaviour of the WAAM-manufactured product. This is crucial for enhancing the characteristics of the product. Furthermore, an inadequate flow rate of shielding gas can have a detrimental impact on the structural integrity for any type of material regardless of any type of heat source employed. This can lead to the formation of voids and porosity owing to chemical reactions with ambient gases [173]. In order to prevent the formation of defects during bead deposition, it is necessary to ensure that an adequate amount of shielding gas is supplied within the WAAM system. This gas acts as a barrier to protect the surrounding area of the weld pool and prevents chemical reactions with atmospheric gases. Doing so effectively prevents the formation of harmful oxides and nitrides [174]. Many researchers have recently implemented innovative methods to optimize shielding gas conditions to emphasise improved deposited material characteristics in WAAM. These methods include using inert chambers or flexible tent shielding to manage turbulence problems, improve surface finish, and improve deposition efficiency (DE) [158,164].

The optimal composition ratio of the shielding gas used in the WAAM process is critical because it affects the heat transfer mechanism during bead deposition [167]. Moreover, it effectively inhibits the occurrence of arc wandering phenomena in extremely reactive substances, such as

Table 3
Relationship matrix between factors affecting the optimization of shielding gas with the final product performance.[103,134,140,142,156–172]

= STRONG, W = WEAK, N = NO EFFECT)

Factors \ Performance	Structural Integrity	Surface Finish	Substrate Warping	Residual Stress	Porosity	Cracks	Hump	Tensile Strength	Hardness	Micro-structure
Type of gas	**S [103], [156], [157]**	**S [158]**	**N**	**N**	**S [159], [160]**	**N**	**N**	**S [161]**	**S [162]**	**S [163]**
Gas flow rate	**S [164], [134]**	**S [164]**	**N**	**N**	**S [159], [142]**	**N**	**N**	**W [165], [134]**	**W [165], [134]**	**N**
Gas composition (ratios and purities)	**S [166], [167]**	**S [168]**	**N**	**N**	**S [162], [140]**	**N**	**N**	**S [169], [168], [170]**	**S [171], [166], [165]**	**S [172], [166]**

titanium and aluminium alloy. Da Silva et al. [175] investigate the effect of varied oxide compositions in argon-based shielding gases of thin aluminum wall structure using GMAW, which successfully optimises the WAAM process by minimising defects formed. Green et al. [170] enhanced the performance of GMAW-WAAM by adjusting the composition ratio of shielding gas for the layer deposition of Grade 91 steel. This allowed them to control the precipitation process and carbonitride morphologies in enhancing mechanical properties of Grade 91 steel at high temperature. In their study, Yamaguchi et al. [103] investigated the issues associated with the metal transfer behaviour of mild steel using GMAW after being subjected to various type of shielding gases. Fig. 11 illustrates the fluctuations in current and voltage seen during the application of 1.17 kJ/cm of heat input for the deposition of five successive layers. The experiment was conducted using Ar gas in case (a) and CO_2 gas in case (b). It has been demonstrated that the type of shielding gases has a substantial impact on the optimization of the WAAM process. N_2 and Ar shielding gases have different impacts on bead formation due to variances in physical and chemical characteristics [163]. The quality of WAAM components can be significantly influenced by the appropriate adjustment of the composition of shielding gases, which in turn affects microstructure characterisation and fracture morphology [162].

2.5. Wire selection

The selection of wire material and its diameter is an important consideration that greatly influences the structural integrity and product quality of NNS WAAM components. The diameter of the wire has a significant impact on the distribution of heat input, deposition rate, and overall quality of the deposited bead. Therefore, controlling the width of the molten pool (WMP) is successfully achieved by optimizing the wire diameter [176]. Varying the wire diameter has a considerable impact on both the breadth and height of the bead [137].

Additionally, implementing numerous wire-feeding mechanisms can enhance material characteristics and control the formation of essential grain structure, grain refining process, and phase transitions [33,177,178]. Thus far, incorporating multiple wire feeding systems has shown to be a very effective and practical solution for addressing material composition issues associated with manufacturing complex geometric components in WAAM [179]. Studies by Feng et al. [180] suggest that the stainless-steel deposition rate of a double-wire feed plasma arc WAAM system may also be improved. Additionally, a more refined microstructure may be produced by employing a dual wire feeding method to enhance the mechanical properties. The presence of two wires results in a notable enhancement in mechanical properties while reducing the number of pores for aluminum alloys using CMT [181,182]. The wire feeding system may be pre-heated to enhance the grain structure, resulting in superior hardness quality [183,184].

Qi et al. [185] developed a dual Wire WAAM system using GTAW to improve the materials characteristics by altering the WFR based on the usage of different types of aluminum wires for layer deposition. Fan et al. [186] developed and verified the feasibility of similar research. Additionally, Somashekara et al. [187] successfully executed a GMAW-based twin-wire weld deposition process using ER70S-6 and ER110S-G filler wires. The mechanical properties, particularly the hardness values, have been effectively enhanced.

Additionally, the selection of wire quality is critical, as any imperfections or defects in the wire will have a consequential impact on the solidification of the deposited beads, thereby influencing the overall strength of the manufactured component. Variation wire size or the presence of cracks and scratches on the wire surface can result in the formation of voids and porosity, which in turn leads to the propagation of cracks inside the deposited bead and layer [188]. By employing high-quality wires, which Murav'ev et al. [189] defined as wires without any surface fractures or scratches, effectively decreases the porosity of the solidified weld pool in molten titanium alloy.

Based on the study conducted in section 2, it can be concluded that the Wire Feed Rate (WFR), Travel Speed (TS), heat input, type of shielding gas, and gas flow rate are the primary process factors in the WAAM technique that have a major impact on the structural integrity of a WAAM structure. A correlation between frequently utilised welding wire materials and specific sets of optimized parameters for the Wire Arc Additive Manufacturing (WAAM) process can be established. In section 5, a radar chart is proposed and developed based on the current trend and research progress in WAAM. The next section will focus on the various deposition strategies utilised in fabricating Wire Arc Additive Manufacturing (WAAM) components. Various methodologies were employed to achieve optimum manufacturing components with superior and excellent structural integrity.

3. Optimization strategy for layer deposition

3.1. Deposition strategy

An essential characteristic of WAAM technology is the ability to achieve a minimal buy-to-fly (BTF) ratio, defined as the mass ratio of the raw materials used to fabricate a component to the final component. BTF for a sliced layer can be calculated by equation (3) [190]:

$$\mathrm{BTF} = \frac{A_{deposited}}{A_{desired}} \tag{3}$$

"$A_{desired}$" refers to the desired area of the shape determined by the sliced CAD model design. On the other hand, "$A_{deposited}$" indicates the actual area that is deposited, which is greatly influenced by the path

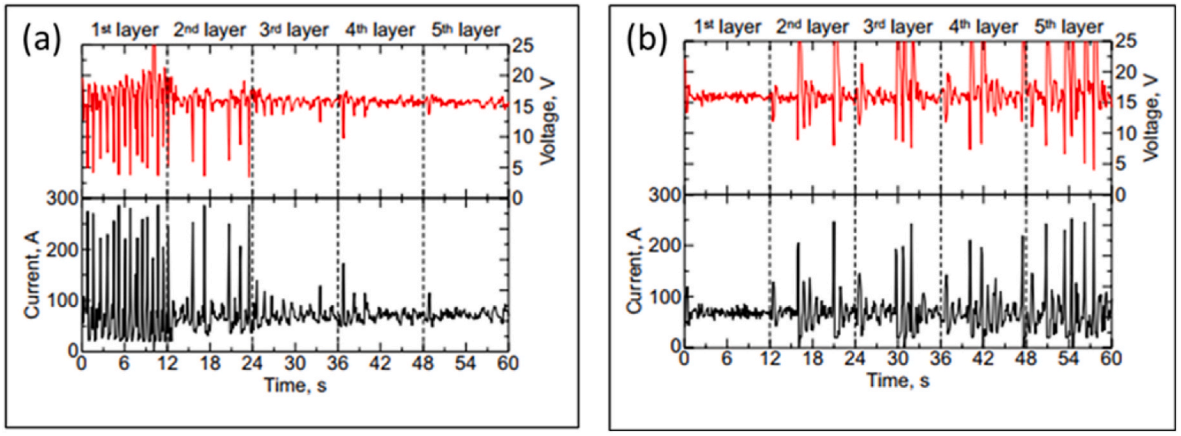


Fig. 11. Variations in current and voltage under different shielding gasses, (a) Ar and (b) CO_2 [103].

planning strategy used. In contemporary industry, a low buy-to-fly ratio is favoured for its ability to streamline the machining process while minimising costs. In order to effectively incorporate such characteristics, it is essential to design an efficient deposition strategy and employ an optimal technique. As illustrated in Fig. 12, deposition strategy is typically divided into the following order: 3D CAD model layer slicing path, the tool-path selection process, and path planning algorithm sequences.

Due to its dynamic nature, the WAAM process faces various challenges and issues during bead deposition. These include varying metal transfer behaviour and heat source use, such as direct crossing, which leads to peaks in thin wall structures. Other challenges include the remelting of previously deposited beads or layers, inaccurate path patterns for opposite angles, and the deformation of beads deposited on overhanging structures. These factors significantly impede the efficiency of the process. As a result, an optimization technique is necessary to accommodate the various deposition strategies implemented [28,66]. Nowadays, much research is being conducted on WAAM in an effort to resolve these issues through the implementation of innovative techniques and methods for path planning, segmentation algorithms, and tool path generation. Three commonly employed deposition strategies for layer-by-layer bead deposition include oscillation, parallel, and weaving, as seen in Fig. 13 [191].

By incorporating multi-axis movement capabilities into additive manufacturing (AM), the process WAAM has become capable of fabricating components without the use of support structures. In order to optimize this capability, a robust and automated algorithm for slicing 3D CAD models in multiple directions has been developed. This algorithm, which includes the Silhouette edges projection algorithm [192], the Transition wall algorithm [193], and the Centroid axis extraction technique [194], efficiently divides the models into layers with minimal need for supportRecently, multi-directional WAAM techniques have been implemented to fabricate complex geometry components with overhang characteristics. This approach is advantageous for industrial applications, allowing for high productivity while minimising capital expenses. The complexity of building an automated system for industrial applications increased due to the interdependence of deposition sequencing and single-layer path planning, making the optimization of the WAAM path planning process more challenging. An essential aspect of the WAAM process is the creation of a path-planning algorithm. This algorithm directs the movement of the deposition tool (weld head) to fill the 2D layers that depict the cross-sectional geometry of the components [195]. Moreover, this algorithm allows for the integration of streamlined work ethics to enhance future process enhancement.

3.2. Tool-path pattern

In order to enhance the diffusion mechanisms of WAAM, it is imperative to incorporate appropriate tools and deposition strategies during the process planning phase [142]. The toolpath must adhere to the geometric characteristics and dimensions of the specified structure while also being consistent with the implemented WAAM methods. Various tool-path patterns have been developed for the implementation of deposition strategies in general for WAAM procedures, as indicated in Table 4. WAAM technology incorporates a multi-directional processing system that can move in the ±X, ±Y, and ±Z directions. Consequently, the configuration of the WAAM system necessitates a distinctive trajectory pattern that allows the automated system to travel in the (X, Y, Z) directions [196].

3.3. Research progress on WAAM deposition strategy

Table 5 presents a summary of available research efforts focused on deposition strategies that have a substantial influence on enhancing the characteristics of final components produced by WAAM. Researchers can utilise specific deposition strategies outlined in Table 5 to enhance the existing characteristics of WAAM structures.

In summary, the substantial detail presented in this section highlights that the deposition strategy significantly impacts the factors that influence the fabrication of WAAM components. Data and findings from various literature sources support this conclusion, demonstrating that slicing and path-planning algorithms are essential components in optimizing the deposition strategy. A significant amount of effort and concepts were effectively implemented in optimizing strategies to address a wide range of issues, allowing for the optimal integration of WAAM results into deposition strategies or the development of path-planning algorithms.

4. Statistical optimization techniques

The optimization and analytical measurement of process parameters in multi-variable WAAM technology are intricate and multifaceted. This complexity arises from the influence of various factors and design challenges on the mechanics of the process. Therefore, achieving an optimal condition necessitates a deep understanding of manipulating multiple combinations of process parameters. Thus, it has become common practice in various manufacturing industry over the last several decades to adopt effective techniques or methodologies, such as statistical analytic calculations, to facilitate this transformational shift [241–245]. Employing such strategies can yield several benefits and contribute to the transformation of Industrial Revolution (IR) 5.0, leading to scientific progress. Statistical analysis is an efficient and successful tool for reducing lead time and capital costs in AM industries. This is because it allows for systematic pre-planning of trial-and-error methods. However, the global investigation of optimizing multi-variable process parameters for high deposition rates about thermo-mechanical and metallurgical characteristics still in progress and continue to developed. As a result, a vast amount of information with varying results has been reported that will be highlighted in this section concisely.

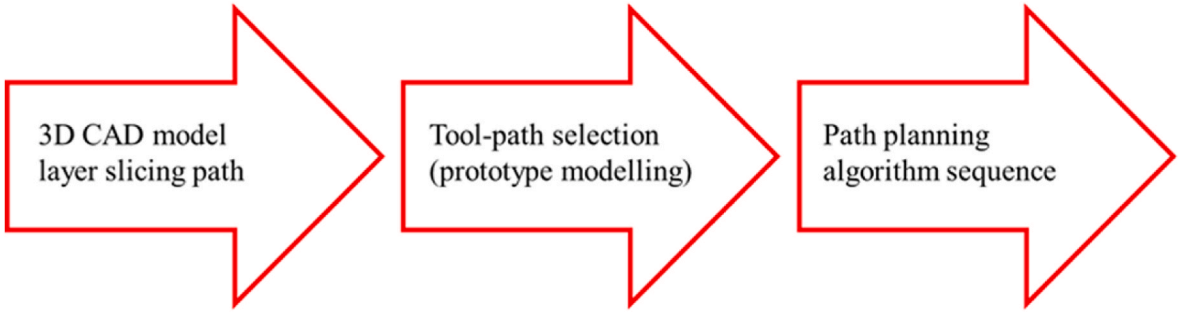


Fig. 12. Process flow in deposition sequence for typical WAAM process.

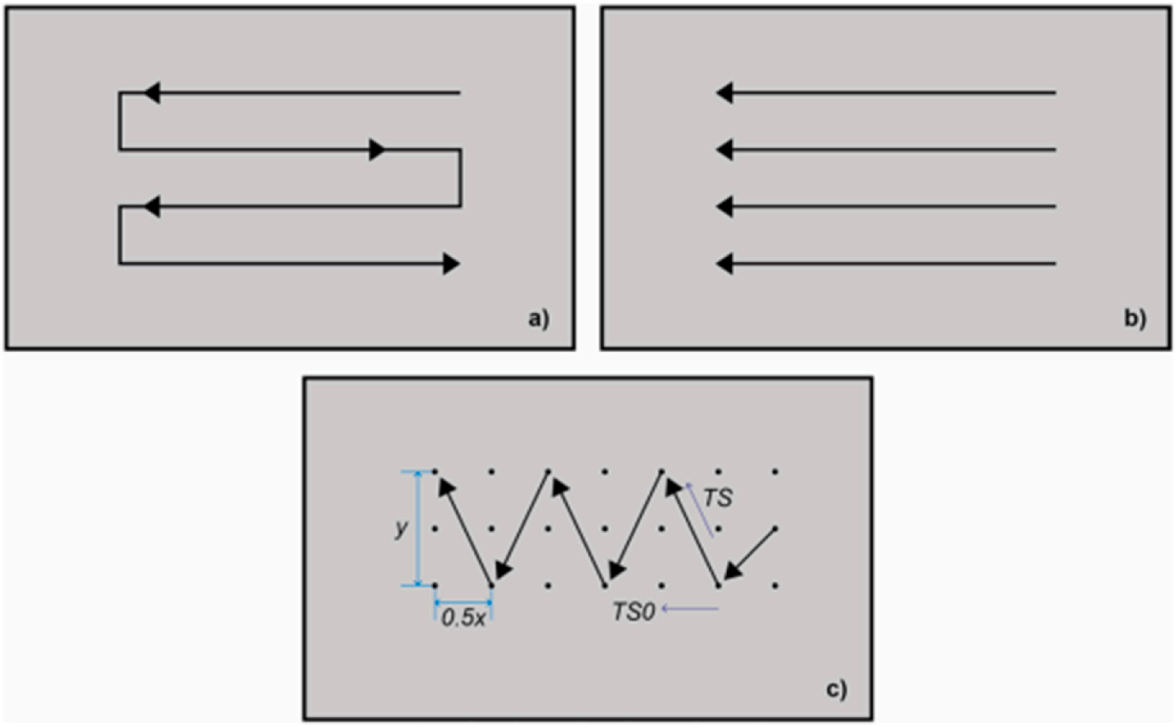


Fig. 13. Deposition strategies: (a) Oscillation, (b) Parallel, (c) Weaving [191].

4.1. Application of statistical tools in WAAM

Researchers have developed and put into practice a variety of evaluation and assessment techniques to evaluate the contribution of process parameters and optimize the WAAM process, as outlined in this section [246]. The research methods utilised in this study consist of Design of Experiments (DoE) techniques, including the Taguchi method and the Full Factorial technique [247], p-values from Analysis of Variance (ANOVA) [248], and regression modelling [249,250]. These techniques were recognised as crucial tools in facilitating the analysis required for process optimization and quality enhancement.

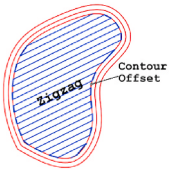
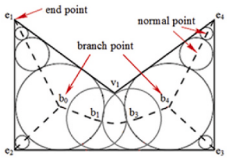
In order to gain a greater understanding of and effectively control the critical challenges of the WAAM process, such as residual stress and deformation, Dinovitzer et al. [10] conducted a set of experiments utilising the L_{16} (4^4) orthogonal array (OA) Taguchi method and one-way ANOVA. The objective was to ascertain the correlation between each factor and response when optimizing material characteristics. This was achieved by manipulating four distinct factors: wire feed rate (mm/min), travel speed (mm/min), current (A), and argon flow rate (CFH). The response obtained showed a notable contribution, indicating that the depth of melt melt-through and surface roughness do not interact with wire feed speed. However, there is a linear rise in bead height.

The process optimization approach is essential for identifying the future potential of WAAM applications. The exceptional focus on height variation as the determining factor poses a challenge in manufacturing WAAM components with consistent structural integrity. Rosli et al. [251] examine the correlation between process parameters and the variation in weld bead height in GMAW-based WAAM wall structures. For this purpose, they employ the Taguchi design with signal-to-noise (S/N) ratio analysis. Likewise, Zavdoveev et al. [252] employed a comparable strategy to develop an efficient welding mode for the pulsed GMAW-based WAAM process. They aimed to increase the deposition rate while maintaining cost-effectiveness. This study presented a method for optimizing the WAAM process by utilising L_9 (3^4) orthogonal arrays (OAs) Taguchi method, and S/N (signal-to-noise) ratio analysis. The objective was to determine the minimum number of experiments required to investigate the impact of each process parameter on the overall WAAM process. As a result of the optimization process, the hardness values increased, leading to greater strength. Additionally, the observed grain structure had undergone improved grain refining.

In their study, Kumar et al. [253] described using a genetic algorithm (GA) to optimize the selection of process parameters for producing NNS components with the dual objectives of minimising void and material waste. GA is an alternative optimization analysis instrument that operates on the principles of stochastic search engine mechanics [254]. In order to forecast the geometry of beads deposited via the WAAM process, models based on response surface methodology (RSM) are constructed using experimental data collected via the Box-Behnken Design (BBD) [255]. The optimization of bead geometry was conducted utilising the response optimizer utility in MINITAB 17 software in conjunction with the desirability function approach. It is possible to fabricate a component with minimal cavity formation and optimum material yield by utilising the optimized process parameters.

Controlling surface morphology and metallurgical behaviour is quite challenging when employing WAAM technology to deposit metals for industrial purposes. Mai et al. [70] examine the optimization of process parameters for thin-walled structures made of 308L stainless steel. They specifically investigate the impact of process stability on mechanical properties and microstructure features. The response of the process parameters used in the experiments resulted in the prediction of the bead geometries, which include the width and height of the beads. The prediction models were developed using RSM and ANOVA. These models were represented by a second-order regression equation. The experimental plan was established using the BBD (Box-Behnken Design) approach. TS has been discovered to have distinct impacts on the shapes of the beads generated. Specifically, the beads' width increases linearly as the applied voltage increases, while the effects on the height of the beads are different. The determined optimal process parameters for solving the multi-objective optimization problems are as follows: current, I = 122 A, voltage, U = 20 V, and TS = 368 mm/min. A. Singh et al. [134] employed a similar method to optimize the factors that impact the width and height of beads in a thick wall structure produced from SS410 wire. These variables include WFR, TS, and gas flow rate. The optimized

Table 4
Summary of WAAM tool-path pattern.

Tool-path pattern	Illustration	References
Raster		[197,198]
Zigzag		[199–201]
Contour		[202–204]
Spiral		[118,205–207]
Continuous		[208–210]
Hybrid		[71,74, 211–214]
Medial Axis Transformation (MAT)		[190,215–218]

parameters are WFR = 5.5 m/min, TS = 63 cm/min, and gas flow rate = 13 liters/min. The WAAM component manufactured does not exhibit any solidification cracking or lack of diffusivity.

In order to optimize the geometry of the weld bead for GMAW-based WAAM of 2.25 Cr-1.0 Mo Steel, a comparable approach was utilised by Vora et al. [256]. The BBD technique and ANOVA were used to conduct experiments and examine the collective effects of numerous process factors, including WFR, TS, and voltage. In addition, the response parameters were further optimized by successfully using the Teaching-Learning-Based Optimization (TLBO) method [257]. In addition, Rosli et al. [258] optimized the process parameters of micro PAW, which include wire feed speed, travel speed, and pulse. Similarly, Banerjee et al. [259] optimized the process parameters of GMAW, which include wire feed speed and travel speed, using the same approaches. Gufran et al. [260] employed a distinct technique to examine the behaviour and characteristics of the weld bead geometry in the GMAW-based WAAM process for low-carbon steel. In order to simplify process planning, they selected the two most important process parameters, current and travel speed, to study the impact of deposition ratio (DR) on the weld bead. They used response surface methodology (RSM) and analysis of variance (ANOVA) to investigate this effect. Regression equations of the second order were effectively formulated and employed to design prediction models for bead geometries. Subsequently, ANOVA is utilised to assess the adequacy of the prediction models.

The process of metal droplet transfer in WAAM technologies is

Table 5
Deposition strategy employed in WAAM (W = Welding Wire, S = Substrate).

Year	Reference	Welding source	Material System	Deposition Strategy
2003	[71]	GMAW	W = 0.8 mm ER70S-6 S═SS308	Higher heat input supplied at the start and gradually reduced until the end of deposition to achieve the intended density and height structure.
2011	[72]	GMAW	W = 1.2 mm H08Mn2Si S = 10 mm S235JR	Use two different deposition directions, (1) same direction and (2) reverse direction where layer deposition in the same direction proved to be better than the reverse deposition sequence.
2012	[73]	CMT	1. W = 0.8 mm ER70S-6 S = 15 mm S355 2. W = 1.2 mm ER4043 S = 12 mm AA6082	Propose an oscillation deposition strategy for inclined wall with angles of 15°, 30°, 45°, and 60° which successfully overcome hump formation.
2014	[74]	GMAW	W = 1.2 mm ER70S-6	The use of a continuous tool-path generation technique, which combines zigzag and contour pattern deposition strategies, results in the formation of a convex polygon structure and leads to enhanced surface accuracy.
2015	[75]	GMAW	W = 1.2 mm ER70S-6	Use two phase Medial Axis Transformation (MAT): phase 1, MAT geometry preparation; and phase 2, path generated from medial axis which improved structure quality and material efficiency.
2015	[219]	GMAW	W = 1.2 mm ER70S-6	Proposed Tangent Overlapping Model (TOM) as a countermeasure to achieve higher multiple bead geometry accuracy whilst reducing material waste optimally.
2016	[220]	GMAW	W = 0.8 mm ER70S-6 S = 12 mm S235JR	Six different deposition strategies have been developed to optimize components with T-crossing features by maintaining height precision for each layer while reducing porosity and residual stress.
2016	[215]	GMAW	W = 1.2 mm ER70S-6	Develop an adaptive Medial Axis Transformation (MAT) path planning algorithm using single bead Artificial Neural Network (ANN) model and multi-bead geometry model base on Tangent Overlapping Model (TOM) to produce void free bead deposition.

(*continued on next page*)

Table 5 (*continued*)

Year	Reference	Welding source	Material System	Deposition Strategy
2016	[221]	GMAW	W = 1.2 mm ER70S-6	Employ Medial Axis Transformation (MAT) path planning algorithm alongside Taguchi method to automate the fabrication of complex geometries.
2017	[222]	GMAW	W = 0.6 mm ER70S-6	Utilise an open-source software, *CuraEngine* to enable the integration of automated slicing of a 3D model to improve slicing function by enhancing bead resolution widths.
2018	[223]	CMT	W = ER4043	Propose a slicing algorithm consist of heuristic method capable of solving complex symmetrical structural features problems.
2018	[224]	GMAW	W = 1.2 mm ER70S-6 S = 10 mm Mild steel	Two distinct path patterns were used to construct an upright rectangular wall structure. The first approach involved welding in a zig pattern with a unidirectional welding direction, while the second way involved welding in a zigzag pattern with alternating welding directions. It was found that the zigzag method had better characteristics compared to the zig method.
2019	[116]	CMT	W = 1.2 mm MARVAL 18S (maraging steel)	Employ oscillation, parallel, and weaving strategies to fabricate 10-layer wall structure with 120 s interlayer cooling time between each deposition in which weaving strategy led to excellent surface finish.
2019	[225]	GMAW	W = 1.2 mm ER4043	Employ three distinct optimization strategies: (1) deposition by weaving, (2) precise control of arc ignition and extinguishing, and (3) localised measurement combined with milling method to minimise variations in layer heights.
2019	[153]	GMAW	W = DSS ER2209 S = 27 mm DSS 2205	Employ two type of deposition strategy: Alternate direction and one direction deposition path to achieve layers with uniform height.
2020	[226]	GMAW	W = 1.2 mm ER70S-6 S = 8 mm S235JR	Employ two deposition techniques: oscillatory and overlapping strategies to fabricate complex geometries, crossing's structure, curvature, and variety of wall thickness.
2020	[227]	GMAW	W = 1.2 mm ER70S-6 S = 8 mm S235JR	Employ control volume concept by comparing between two different deposition strategy: oscillatory and overlapping strategies to minimise defects.
2020	[60]	CMT	W = 1.2 mm ER120S-G S = 10 mm S355	Employ oscillation and parallel deposition strategies to produce single bead welds and wall geometries which reveal lower yield strength is achievable.
2020	[228]	CMT	W = 1.2 mm SS316L S = 15 mm SS316L	Employ weaving strategy whilst increasing the torch angle yield excellent surface morphology alongside grain refinement.
2021	[229]	CMT	W = 1.2 mm ER2319 S = 12.7 mm AA6082-T6	Utilise oscillation and parallel deposition techniques to construct a thicker segment of a linear wall in order to enhance its hardness.
2022	[191]	CMT	W = 1.2 mm ER70S-6 S = 20 mm Carbon steel plate	Perform oscillation, parallel, and weaving strategies to fabricate walls, blocks, and bimetal structure.
2022	[57]	CMT	W = 1.6 mm DSS ER2209 S = 10 mm Q235A steel	Two additive paths were used to deposit 10-layer wall structure, unidirectional and reciprocating addition.
2022	[230]	GMAW	W = 1.2 mm ER70S-6 S═S235JR	Three distinct deposition strategies were utilised to build a wall structure with an inclined angle, specifically for a particular component dimension. Three different strategies are being tested: (1) the go strategy, (2) the go strategy with changed entrance and exit conditions, and (3) the back-and-forth method.
2022	[56]	GMAW	W = 1.2 mm FCWA-AM Super Duplex Stainless Steel (SDSS)	Different process parameter were applied for each nth layer to achieve austenite ratio of 45%.
2022	[131]	GMAW & PAW	W = 1.2 mm ER70S-6 and ASS 316L S = 10 mm ASS 316L	Use two different deposition techniques: superimposed and overlapped strategy to deposit defects-free bimetallic wall structure.
2022	[231]	GTAW	W = 1.0 mm ASS 316L (ER 316LSi) and Inconel 625 (ER NiCrMo-3) S = 6 mm Mild steel plate	FGMs properties were compared by employing two different deposition strategies: direct interface and smooth transition where direct interface has proven to be a superior strategy in terms of strength and elongations upon failure.

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Table 5 (*continued*)

Year	Reference	Welding source	Material System	Deposition Strategy
2022	[232]	CMT	W = 1.2 mm Ti–6Al–4V S = 4 mm Ti–6Al–4V	Employ zigzag deposition strategy on different condition of CMT transfer modes to achieve stable structure properties.
2022	[233]	PAAM	W = 1.2 mm Ti–6Al–4V S = 4 mm Ti–6Al–4V	Compare two sets of PAAM specimen: (1) layer deposition without rolling force, (2) layer deposition with in-situ rolling force of 15 kN.
2022	[234]	CMT	W = 0.9 mm ER70S-6 S = 10 mm Q235 steel plate	Suggest an innovative approach for automating the path planning process in robotics by including an Automated Robot Offline Programming (AOLP) engine. This approach uses a collision matrix as a heuristic strategy to simplify the complexity of the path-planning process.
2022	[235]	CMT	W = Kiswel 1.2 mm M-308 S = 25 mm SS304	Integrating a regression model with a Support Vector Machine (SVM) classifier can diminish the variation in bead deposition between the arc strike zone and the intermediate zone.
2022	[236]	CMT	W = 1.2 mm ER4043 S = 3.15 mm AA6061-T6	Employ two different path planning strategy, (1) Uni-directional, and (2) Bi-directional in reducing heat input.
2022	[237]	CMT	W = 1.2 mm Aluminum ER2319 S = 10 mm AA6082-T6	Utilise two deposition techniques for torch movement, namely the Hatching strategy and the Circling method, while alternating the travel direction from odd to even layers to ensure stability.
2022	[238]	PAW	W = 1.2 mm ASS 316L-Si S = 15 mm S235JR	Optimal high-quality X-Cross intersection geometry's structure were obtained through the employment of both Cross-waving and Cross-Overlapping deposition strategies.
2023	[239]	CMT	W = 1.2 mm Low alloy steel (Fe–2Mn–1Si-0.08C) S = 10 mm Low carbon steel (Fe-1.4Mn-0.35Si-0.2C)	Multi-pass paths with varying center distances and extra gap-fill paths have been employed through the utilization of Artificial Neural Network (ANN) to produce gap-free components.
2023	[240]	CMT	W = 1.2 mm Aluminum ER2319 S = 10 mm 2Al4 Aluminum alloy	Spiral and asymmetrical trapezoid arc oscillation deposition pattern employed allowed the reduction of pores formation from 2.64% to 1.09% and 1.22% respectively.

complicated due to the involvement of several unknown factors, including electromagnetic force (EMF), tension force, gravity, and the weight impact of the droplet. Liang et al. [261] conducted a study to evaluate the dynamic behaviour of metal transfer in the form of droplets under different transfer modalities. They used ANOVA to determine the importance of each parameter involved. The presence of defects and imperfections, such as voids and porosity, in Aluminum-based WAAM components, highlights the need to include statistical analysis in AM research. By controlling the size and positioning of the porosity, it is possible to achieve an optimal structure with excellent quality using this manufacturing technique. Derekar et al. [104] conducted a statistical study using ANOVA to examine the variations in porosity diameter in several specimens fabricated using two distinct metal deposition circumstances, namely pulsed MIG and CMT. The investigation focused on verifying these differences by examining the p-values.

In a study conducted by Naveen Srivinas et al. [262], optimization research was carried out to fabricate a thin plate component using the aluminum-based WAAM technique. The study employed the Grey Relational Analysis (GRA) methodology, which involved the following essential tasks: normalising the experimental data, calculating the grey relational coefficients (GRC), and determining the grey relational grades (GRG) for each run. In addition, this study utilised the L9 OA Taguchi method and ANOVA to minimise the number of trials and examine the effects of various input process variables (factors) - wire feed rate (WFR), gas flow rate, and travel speed - on the output variables (response), specifically the deposited weld bead geometry. These variables were previously suggested by Manikandan et al. [263] for the electrochemical drilling process. The methodology recommended for this study is depicted in Fig. 14. Based on the study conducted, the combination of process parameters with a WFR of 7.5 m/min, a traverse speed (TS) of 0.6 m/min, and a gas flow rate of 18 L/min is determined to be the most optimized for maintaining a high-quality thin plate structure.

Multi-variable process parameters necessitate a unique optimization method in order to enhance productivity. As a result, Lee [19] developed a design for optimizing process parameters in the WAAM technique using a non-parametric method called Gaussian Process Regression (GPR) modelling. This approach accurately calculates minor effects and uncertainties, formulating a unique WAAM optimization model. The model achieves higher productivity standards, improved structural integrity, and higher quality.

In summary, researchers may streamline the process of optimizing WAAM by utilising appropriate statistical analysis methods as tabulated

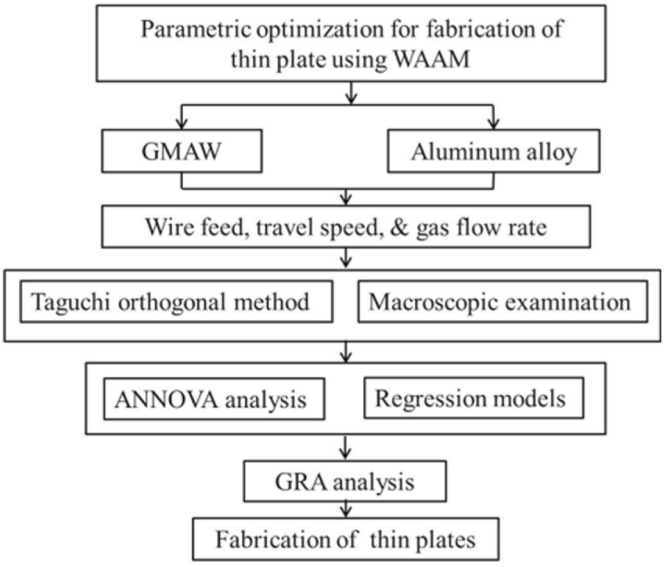


Fig. 14. Proposed research methodology flowchart [262].

in Table 6 which led to enhanced overall WAAM productivity. In order to do this, researchers must endeavour to determine the appropriate process parameter control that has a substantial impact on the result of the desired WAAM structure. Subsequently, an appropriate statistical analysis will be used to minimise errors related to the WAAM process.

5. Recommendations for an optimized WAAM process

WAAM possesses immense potential as the future forefront technology for the additive manufacturing sector. The growing demand for the additive manufacturing (AM) sector, whether for educational or commercial purposes, compelled WAAM technology to develop near-net-shape (NNS) components of superior quality while reducing lead time and capital expenditure. Optimization approaches and innovations are increasingly essential to minimise waste and uphold overall process efficiency. Moreover, implementing optimization techniques has the potential to enhance sustainability by increasing productivity in aiding the economic growth of a nation. Upon conducting a thorough review of the reported study, a radar chart is suggested and developed to establish a correlation between the wire used and the process parameters of the WAAM technique. In addition to the suggested radar chart, a framework for an optimization-based algorithm is proposed.

Table 6
Summary on the characteristics of various analytical optimization methods.

Analytical Optimization Methods	Characteristics	Reference
Taguchi	-Simplify the WAAM process by focusing on reducing the variations of parameters selected. -A robust design to save time and cost whilst improving the quality of deposited structure.	[251,252,262,264,265]
Full Factorial	-Understand how each WAAM process parameter affects the structural quality of deposited parts. -This method considers all possible combinations of each process parameter involved in identifying which have the most significant impact on the WAAM process. -Investigate the interaction of each process parameters with each other.	[65,266–269]
Response Surface Methodology (RSM)	-Simplify the WAAM process by determine the best settings for each process parameter involved while ensuring high quality structure with minimal error. -Utilise mathematical modelling in designing the experiments which is more complex than Taguchi and Full Factorial as it requires high level of expertise to achieve high level of precision in optimizing WAAM process. -Less time and resources needed to execute a precise WAAM optimization.	[70,112,253,256,258–261,270]
Analysis of Variance (ANOVA)	-Identify the important process parameters that affects the deposition. -Quantifying how each process parameter contributes towards the WAAM process. -Employing Design of Experiments (DoE) to collect the response data systematically, such as tensile strength, hardness, etc. -Validating the statistical models obtained for accurate and precise optimization analysis.	[70,112,251–253,256,258–262,264,266,270]
Regression Modelling	-Provide a prediction model on how the changes in process parameters affect the quality of WAAM structure. -Quantifying relationships between each input (process parameters) and output (response). -Quantitatively identifies the significant factors that influenced the overall WAAM performance.	[70,112,256,258–262,266,270]

5.1. Proposed radar chart

By making the appropriate selection between wire material and process parameter using this chart; one can arrive at an optimal decision making for a preliminary WAAM configuration setup specifically for the layer deposition of wall structure application, as seen in Fig. 15. Other application stated in Jafari et al. [271] such as overhang features, crossing feature, corner junction, cylindrical junction, lattice, and struts may also be included as the deposition process parameter employed fall between the proposed range as shown in Fig. 15 and Table 7. The scale depicted on the chart is converted into the specific values of process parameters listed in Table 7. Discovering the most optimal process parameter chosen for a preliminary WAAM configuration is crucial. The WAAM process regulates heat input by selecting proper voltage (V) and/or current (A) values. For bi-metal wire material system, the ideal result may be achieved by combining specific process parameters for each specified material. At the same time, the chart does not specify the precise success rate of a WAAM setup due to factors such as the type of heat source, wire feeding mechanism, and material system; the chosen values can serve as foundations for selecting an optimal process parameters.

5.2. Proposed framework

As depicted in Fig. 16, the framework proposed are applicable to be integrated with any type of material and heat source used. This framework start with the selection of WAAM process parameter and end with performing tests and analysis to procure an ideal properties for the final component in resembling a Near-Net-Shape product. After process parameter selection, three distinct optimization strategies are employed and defects analytical measurement were performed to ensure a successful bead or layer deposition. Then, microstructural characterization and quality control were conducted to identify the structural integrity of the WAAM parts. Proper post processing treatment will be selected for further quality improvement. If the quality improvement were successful, it will proceed to final component properties testing and if it did not meet the required specifications desired, the algorithm will proceed

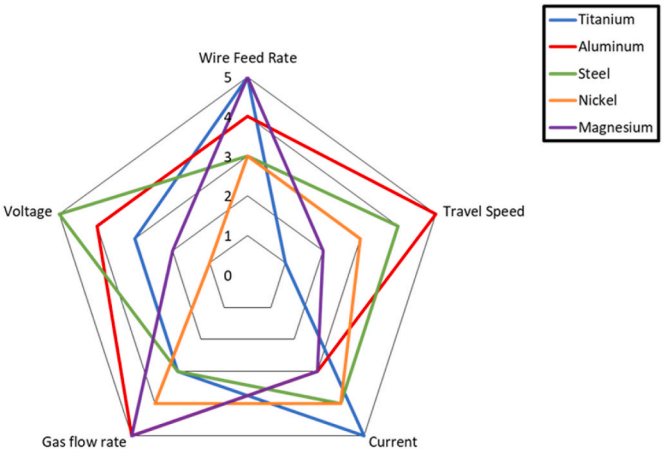


Fig. 15. Proposed optimization correlation chart between type of wire materials and WAAM process parameter.

Table 7
Range of process parameter values correspond towards scale of the correlation chart.

Scale	Wire feed rate (m/min)	Travel speed (m/min)	Voltage (V)	Current (A)	Gas flow rate (L/min)
1	–	0.1–0.6	12–14	–	–
2	–	0.2–0.8	10–17	–	–
3	1–7	0.3–0.8	14–17	70–180	10–20
4	2–8	0.1–1.1	11–19	100–230	15–20
5	1–10	0.3–1.2	14–27	140–350	15–25

to process parameter selection instead.

The algorithm proposed would significantly enhance the optimization of algorithmic systems, leading to significant advancements in the future development of the WAAM optimization process for various materials with complex geometrical characteristics. Optimal filling process algorithm may consist of enhanced slicing 3D CAD model, improved path planning directions, effective process parameters selection procedures, and high-quality bead deposition strategy mechanism. Enhancements may be made to slicing and path planning algorithms by minimising the overall length of the routes, minimising the number of intersections between sub-paths, and minimising the route accessible for certain structural elements. These improvements aim to save time and minimise material waste. Prioritising the ease of integrating the proposed algorithm with current WAAM equipment and its adaptability is prioritised. This would significantly save non-value-added time and facilitate the development of essential elements for industrial adoption.

Utilising configurable algorithms for deposition techniques provides limitless adaptability, enabling the improvement of material quality and WAAM setups under various environmental and process circumstances. Hence, it is imperative to design innovative optimization algorithms that effectively balance the slicing and path planning algorithm, process parameter and heat input choices, material qualities, and complex geometrical aspects to get the optimal result.

6. Future insight of WAAM optimization process

The investigation of various heat sources utilised in the WAAM process is regarded as a critical area of research in the AM industry due to its adaptability in processing a wide range of materials. In the future, it may be possible to incorporate multiple heat sources into a WAAM system configuration to optimize heat distribution, temperature gradient, and various types of thermal cycles, thereby enhancing mechanical performance and microstructure. An extensive comprehension of material attributes results in an optimal process configuration, influencing the selection of optimal process parameters.

Utilising an improved measurement control approach may effectively enhance the structural integrity and quality of WAAM components, serving as a valuable tool for optimization. Moreover, the credibility of the optimization process might be enhanced by integrating diverse approaches to analysis and testing. This encompasses the integration of experimental data and statistical analysis. Using various statistical analysis tools allows for the development of more precise data, which is essential for optimization. In addition, the integration of optimization analysis may effectively save time and cost by eliminating the need for trial-and-error approaches, hence minimising material waste. By incorporating the topology optimization (TO) technique into future WAAM systems, it becomes possible to construct a geometric structure that restructures the lattice parameters to reduce material waste. This restructuring allows for weight reduction without compromising the mechanical properties and quality of the components. In order to efficiently develop resource-feasible optimization processes, it is recommended to utilise advanced simulation computation and mathematical modelling tools such as machine learning, decision science heuristic approach, and process modelling design. These tools are beneficial because they can handle a wide range of potential process combinations.

Integrating several manufacturing methods might improve components' characteristics over time and reduce overall costs by eliminating the difficulties related to WAAM [272]. For instance, the integration of the WAAM and synchronous electromagnetic stirring (EMS) production system enables the enhancement of both grain refinement and tensile shear strength [273]. In addition, using a multi-sensor monitoring system and intelligent control, such as Artificial Intelligence (AI), in the WAAM setup allows for more reliable process optimization by effectively reducing and minimising errors [223]. In order to compensate WAAM manufacturability issues involving complex and inaccessible path geometries with difficult offset control, a method based on a data-driven Artificial Neural Network (ANN) system can be implemented to obtain optimized set of solutions.

7. Conclusion

A comprehensive analysis of the Wire Arc Additive Manufacturing (WAAM) process's optimization process has been effectively presented. The analysis primarily focuses on the selection procedure for process parameters, deposition strategies (including path planning algorithms), and the applicability of statistical analysis tools. By thoroughly examining the material's functionality and the performance characteristics of the WAAM-manufactured components, several suggestions have been put forth to advance and improve the optimization method for producing NNS components that are of superior quality and free from defects. WAAM has consistently demonstrated its superiority as the primary production method and is widely utilised in industrial applications for metal Additive production (AM) technology. This manufacturing technique is characterised by a high deposition rate, making it adaptable and advantageous for monetization and commercialization. The deposition rate employed is determined by two primary parameters: wire feed rate (WFR) and heat input. Various heat sources possess distinct benefits relative to one another. Understanding different materials' varying heat input and process control needs is essential in WAAM. In summary, WAAM is a viable alternative to conventional manufacturing techniques on account of its superior mechanical and microstructural properties and its ability to facilitate applications across multiple disciplines. However, further investigation remains essential and required in order to achieve optimization objectives.

Author contribution

All the authors were ranked according to their contribution to the article.

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Data availability

Not applicable.

Code availability

Not applicable.

Ethics approval

The authors declare that the ethics of this article is approved.

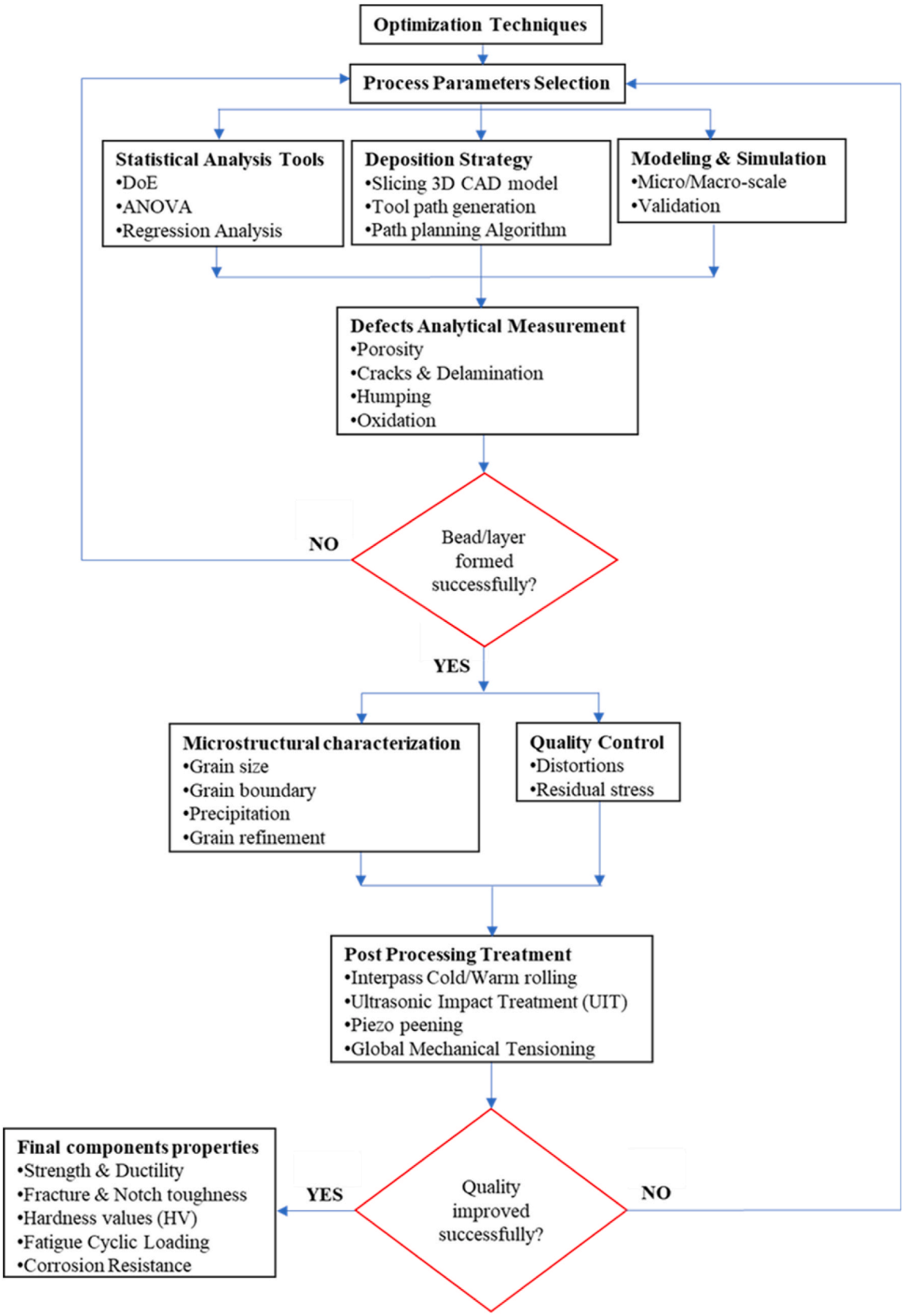


Fig. 16. Optimization-based algorithm framework for optimal WAAM process.

Consent to participate

All the authors agree that the editorial board is qualified to examine the article.

Consent for publication

All the authors agree that the editorial board may publish the article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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